MIL-R-47004(MI)
19 April 1974
SUPERSEDING
MIS-14380
7 July 1967

MILITARY SPECIFICATION

RESIN, EPOXY BASE, THIXOTROPIC

This specification is approved for use by all departments and agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for a thixotropic epoxy base resin.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

Federal

PPP-C-96	Can, Metal, 28 Gage and Lighter
PPP-P-704	Pails, Shipping Steel (1 Through 12 Gallon)

Military

MIL-I-16923	Insulating Compound, Electrical, Embedding

STANDARDS

Federal

FED-STD-406	Plastics, Methods of Testing

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Military

MIL-STD-129 Marking for Shipment and Storage

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials

ASTM D 149	Dielectric Breakdown Voltage and Dielectric Strength of Electrical Insulating Materials at Commercial Power Frequencies, Tests For
ASTM D 648	Deflection Temperature of Plastics Under Flexural Load, Test For
ASTM D 792	Specific Gravity and Density of Plastics by Displacement, Tests For

(Applications for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19102.)

OFFICIAL CLASSIFICATION COMMITTEE

Uniform Freight Carrier Classification Rules

(Application for copies should be addressed to the Official Classification Committee, 1 Park Avenue at 33rd Street, New York, New York 10016.)

Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.

3. REQUIREMENTS

3.1 Preproduction sample. Unless otherwise specified (see 6.2), a preproduction sample of epoxy base resin shall meet the requirements of this specification.

3.2 Material. The thixotropic resin shall consist of an epoxy

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base resin and a compatible curing agent capable of being fully cured at a temperature of not more than plus 95 degrees Celsius (C) (plus 203 degrees Fahrenheit (F)) for 5 hours plus or minus 1 hour, with a post cure at a temperature of not more than plus 120 degrees C (plus 248 degrees F) for 2 hours plus or minus 15 minutes.

3.3 Electrical properties.

3.3.1 Dielectric strength. The dielectric strength of the cured epoxy resin shall be not less than 300 volts per mil.

3.3.2 Volume resistivity. The volume resistivity of the cured epoxy resin at plus 23 degrees C (plus 73.4 degrees F) and 50 plus or minus 2 percent relative humidity shall be not less than 1 times 10^{14} ohm-centimeters.

3.4 Mechanical properties.

3.4.1 Hardness. The cured epoxy resin shall have a Barcol hardness of 45 plus or minus 10 with a 10 second reading.

3.4.2 Heat distortion temperature. The heat distortion temperature of the cured epoxy resin shall be not less than plus 115 degrees C (plus 239 degrees F).

3.4.3 Thermal shrinkage. The linear thermal shrinkage of the cured epoxy resin shall be not more than one percent.

3.4.4 Thermal conductivity. The thermal conductivity of the cured epoxy resin shall be not less than 8 times 10 to the minus 4 calories per second per square centimeter per degree Celsius per centimeter.

3.4.5 Thermal shock. The cured epoxy resin shall withstand not less than 10 cycles of thermal shock at the low temperature of minus 55 plus or minus 2 degrees C (minus 67 plus or minus 3.6 degrees F) and the high temperature of plus 130 plus or minus 3 degrees C (plus 266 plus or minus 5.4 degrees F).

3.5 Physical properties.

3.5.1 Specific gravity. The specific gravity of the cured epoxy resin shall be 1.53 plus or minus 0.05.

3.5.2 Shelf life. The shelf life of the base resin and the hardener in unopened containers shall be not less than 12 months from the date of manufacture when stored at a temperature not more than plus 27 degrees C (plus 80.6 degrees F).

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3.5.3 Contamination. The base resin and hardener shall be free of dirt, foreign materials, and other contamination.

3.5.4 Application life. The application life of the mixed resin system shall be not less than 72 hours when used and stored at not more than plus 23 degrees C (plus 73.4 degrees F).

3.5.5 Workmanship. The workmanship shall be such as to insure a high quality product which is uniform and in conformance to this speicfication.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Preproduction sample. The preproduction sample shall be prepared using the same methods proposed for the preparation of subsequent production lots of material. Preproduction samples which do not meet all the requirements of this specification shall be subject to rejection and returned to the supplier.

4.3 Classification of tests. The examination and testing shall be classified as follows:

a. Preproduction tests.

b. Qualtiy conformance tests.

4.3.1 Preproduction tests. Preproduction tests shall be conducted only on the preproduction sample and shall consist of all the examinations and tests specified herein.

4.3.2 Quality conformance tests. Quality conformance tests for acceptance of the base resin and hardener shall consist of the following examinations and tests:

a. Hardness.

b. Specific gravity.

c. Contamination.

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4.3.2.1 Lot size. Lot size shall consist of all the base resin and hardener manufactured at one time, from one batch, forming part of one contract or order, and submitted at the same time and place.

4.3.2.2 Sampling. Unless otherwise specified (see 6.2), 1/2 pint of base resin and 1/2 pint of hardener from 1 unit representative of each lot shall be selected at random for quality conformance testing. Each packaged container shall be considered as a unit of product. Failure of sample to meet the quality conformance requirements shall be cause for lot rejection.

4.4 Test methods.

4.4.1 Electrical properties.

4.4.1.1 Dielectric strength.

4.4.1.1.1 Test specimens. The test specimen size for the dielectric strength test shall be 3 by 3 by 0.125 plus or minus 0.010 inches. The base resin and hardener shall be mixed in accordance with the manufacturer's instructions and shall be cured for a duration of 3 hours at a temperature of plus 121 degrees C (plus 250 degrees F).

4.4.1.1.2 Procedure. The test shall be performed in accordance with ASTM D 149. The specimen shall be immersed at least 1 inch below the surface of a bath of high grade transformer oil. The rate of voltage rise shall be 500 volts per second. The breakdown voltage shall be determined and the dielectric strength in volts per mil shall be calculated. The dielectric strength shall be as specified in 3.3.1.

4.4.1.2 Volume resistivity. The volume resistivity of the cured epoxy resin shall be determined in accordance with MIL-I-16923. The volume resistivity shall be as specified in 3.3.2.

4.4.2 Mechanical properties.

4.4.2.1 Hardness. The Barcol hardness of the cured epoxy resin shall be determined by direct reading of Barcol impressor. Four determinations ahll be made and the average recorded. The Barcol hardness shall be as specified in 3.4.1.

4.4.2.2 Heat distortion temperature. The heat distortion temperature of the cured epoxy resin shall be determined in accordance with ASTM D 648. The heat distortion temperature shall be as specified in 3.4.2.

4.4.2.3 Thermal shrinkage. The linear thermal shrinkage of the cured epoxy resin shall be determined in accordance with FED-STD-406, Method 2031. The thermal shrinkage shall be as specified in 3.4.3.

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4.4.2.4 Thermal conductivity. The thermal conductivity of the cured epoxy resin shall be determined in accordance with MIL-I-16923. The thermal conductivity shall be as specified in 3.4.4.

4.4.2.5 Thermal shock. The thermal shock of the cured epoxy resin shall be determined in accordance with MIL-I-16923. The thermal shock shall be as specified in 3.4.5.

4.4.3 Physical properties.

4.4.3.1 Specific gravity. The specific gravity of the cured epoxy resin shall be determined in accordance with ASTM D 792, Method A. The specific gravity shall be as specified in 3.5.1.

4.4.3.2 Shelf life. The epoxy base resin and hardener shall be tested for the requirements specified herein after completion of storage period (see 3.5.2). The manufacturer's certificate of compliance with shelf life requirements may be accepted during the storage period.

4.4.3.3 Contamination. The contamination of the materials shall be determined visually. The contamination shall be as specified in 3.5.3.

4.4.3.4 Application life. The application life of the mixed resin system shall be determined visually. The application life shall be as specified in 3.5.4.

4.5 Preservation, packaging, packing, and marking. The preservation, packaging, packing, and marking shall be examined to determine compliance with Section 5.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, packing, and marking. Unless otherwise specified (see 6.2), preservation, packaging, packing, and marking shall be as specified herein.

5.2 Preservation and packaging. Preservation and packaging shall be Level A or C as specified (see 6.2).

5.2.1 Level A.

5.2.1.1 Unit packaging. The epoxy resin and curing agent shall each be packaged in its own container. The epoxy resin shall be packaged in tin plate containers with metal lids conforming to PP-C-96, Type V,

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Class 2 or drums conforming to PPP-P-704, Type II, as specified (see 6.2). The curing agent shall be furnished in glass bottles or equal. Containers shall be nonreactive to the contents. The size of the container to be specified by the procuring activity (see 6.2). The epoxy resin and curing agent shall be packaged in such a manner that deterioration or physical damage during shipment and storage is prevented.

5.2.2 Level C. Unit package shall be in accordance with 5.2.1.1. Unit containers shall be in accordance with the supplier's commercial practice. Packages shall be protected against deterioration or physical damage during shipment from the supply source to the first receiving activity for immediate use.

5.3 Packing. Packaging shall be Level A, B, or C as specified (see 6.2).

5.3.1 Level A. Units packaged in accordance with 5.2.1.1 shall be packed to conform to the overseas shipment requirements of PPP-C-96, Appendix. Pails over 5 gallon will require no overpacking.

5.3.2 Level B. Same as Level A.

5.3.3 Level C. Packing shall be in accordance with the Uniform Freight Carrier Classification Rules or other carrier regulations as applicable to the mode of transportation.

5.4 Marking. In addition to any special marking required by the contract or purchase order (see 6.2), unit packages, intermediate packages, shipping containers shall be marked in accordance with the requirements of MIL-STD-129.

6. NOTES

6.1 Intended use. The epoxy resin covered by this specification is intended for use as an encapsulant for welded electronic circuits.

6.2 Ordering data. Procurement of documents should specify the following:

a. Title, number, and date of this specification.

b. Whether a preproduction sample is required, and, if so, pertinent details (see 3.1).

c. Sampling plan if other than specified (see 4.3.2.2).

d. Applicable level of preservation, packaging, and packing (see 5.2 and 5.3).

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e. Marking requirements if different (see 5.4).

f. Quantity required.

g. Requirements for preservation, packagin, packing, and marking, if different (see 5.1).

h. Type and size of container required (see 5.2.1 1).

6.3 Supersession data. This specification includes the requirements of Missile Interim Specification MIS-14380, dated 7 July 1967.

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